

PROGRAMMING IN C: COLLABORATIVE ASSIGNMENT REPORT:

TEAM MEMBERS:

DISHANTH VASUPAL K R - RVUN25CSE364

FRANCIS REUBEN R - RVUN25CSE408

G HARIKRISHNAN - RVUN25CSE045



PROGRAM:

```
metanaux@metanaux (192.168.64.9) - byobu
GNU nano 7.2 Prog2.c
#include <stdio.h>
int fact(int x){
    return x == 0 ? 1 : x * fact(x - 1);}
void main(){
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    printf("Factorial of %d is %d\n", n, fact(n));}
```

OUTPUTS:

n = 0

```
metanaux@metanaux (192.168.64.9) - byobu
[0.007s] metanaux@metanaux:~/Frank/PiC$ ./Prog2
Enter a number: 0
Factorial of 0 is 1
[1.311s] 80 metanaux@metanaux:~/Frank/PiC$
```

n = 6

```
metanaux@metanaux (192.168.64.9) - byobu
[0.006s] metanaux@metanaux:~/Frank/PiC$ ./Prog2
Enter a number: 6
Factorial of 6 is 720
[2.223s] 80 metanaux@metanaux:~/Frank/PiC$
```

n = 10

```
metanaux@metanaux (192.168.64.9) - byobu
[0.004s] metanaux@metanaux:~/Frank/PiC$ ./Prog2
Enter a number: 10
Factorial of 10 is 3628800
[1.242s] 80 metanaux@metanaux:~/Frank/PiC$
```

ALGORITHM:

Algorithm: Factorial of a Number (Using Recursion)

1. Start
2. Declare an integer variable n.
3. Define a function fact (x) that:
 - If $x == 0$, return 1.
 - Else, return $x * \text{fact}(x - 1)$.
4. In the main function:
 1. Display the message "Enter a number: ".
 2. Read the input value and store it in n.
 3. Call the function fact (n) and store/print the result.
 4. Display the message "Factorial of n is result".
5. Stop

FLOWCHART:

